EXP 8

Data Visualization Using Tableau

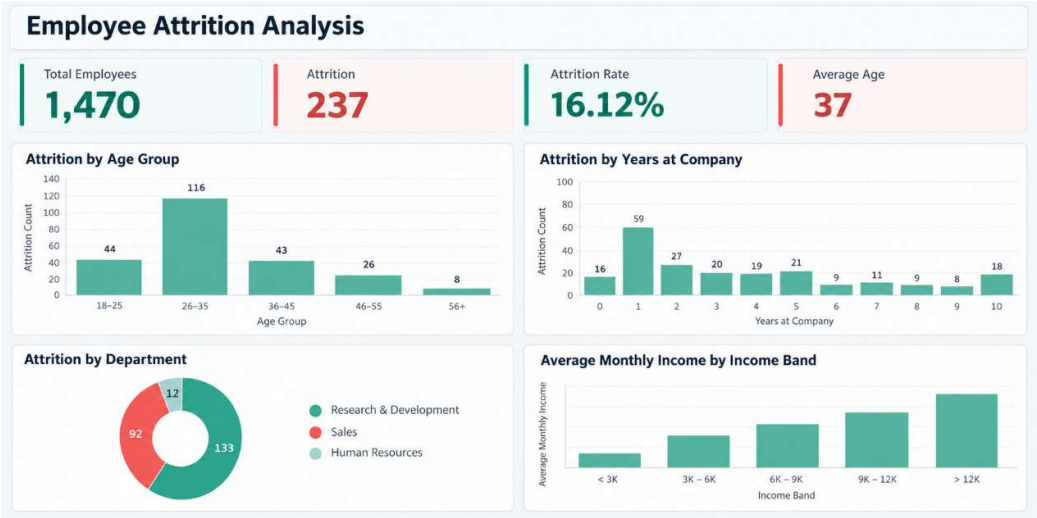
AIM:

To learn the Tableau Desktop workflow by connecting to employee attrition data, cleaning and preparing the data, creating visualizations such as bar charts, a donut chart and KPI cards, and arranging them into an interactive employee attrition dashboard.

ALGORITHM:

1. Open Tableau Desktop and identify the Data pane, Analytics pane, worksheet area, Show Me panel and Dashboard workspace.
2. Click Connect → To a File → Text File, select the employee attrition CSV dataset and connect to it.
3. Open the Data Source page and verify the imported data. Check for missing values, duplicate records and incorrect data types.
4. Verify that fields such as Department, Age, Years at Company, Monthly Income and Attrition have the appropriate data types, and create calculated fields where required.
5. Open a new worksheet and create KPI cards displaying Total Employees (1,470), Total Attrition (237), Attrition Rate (16.12%) and Average Age (37).
6. Create a bar chart titled Attrition by Age Group to compare the number of employees who left across different age groups.
7. Create a column chart titled Attrition by Years at Company to analyse employee attrition across different years of service.
8. Create a donut chart titled Attrition by Department to display the attrition counts for Research & Development (133), Sales (92) and Human Resources (12).
9. Create a column chart titled Average Monthly Income by Income Band to compare average monthly income across the five income bands.
10. Open the Dashboard workspace, arrange the four KPI cards and four charts neatly, and apply consistent colours, titles, labels, legends and formatting.
11. Add relevant filters and configure dashboard actions where required, verify the visualizations and displayed values, and save the completed Tableau workbook as a .twb or .twbx file.

OUTPUT:



RESULT:

The Employee Attrition Analysis dashboard was successfully created in Tableau, displaying key KPIs and visualizations to analyse employee attrition by age, years at company, department and income band.